SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 30 Minutes
QUESTIONS: 2 **Total Marks:20**

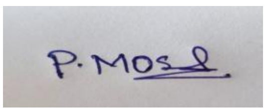
Annexure A

<u>Descriptive Written Test</u>

Code of Conduct:

I P.Moshiya(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO1 – ASSESSMENT TOOL 3

Descriptive Written Test

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

QUESTIONS: 2 Duration: 45 Minutes **Total Marks: 20**

Criteria	Weightage	Q1 (10)	Q2 (10)	Total Marks (20)
Understanding of Concepts	25% (2.5)	/2.5	/2.5	/5
Explanation	25% (2.5)	/2.5	/2.5	/5
Application	20% (2)	/2	/2	/4
Organization	15% (1.5)	/1.5	/1.5	/3
Accuracy	10% (1)	/1	/1	/2
Clarity	5% (0.5)	/0.5	/0.5	/1
Total per Question	**10 Marks**	**/10**	**/10**	**/20**

100

1. Smart Home Automation System – Intelligent Agents

Introduction

A smart home automation system controls lighting, temperature, and security by monitoring environmental conditions and user activities. Intelligent agents help the system make decisions based on sensor inputs, past experiences, and user preferences.

Types of Intelligent Agents

a) Simple Reflex Agent

A **Simple Reflex Agent** makes decisions only based on the current percept (input) using predefined condition-action rules.

Working in Smart Home:

- If motion is detected → Turn on lights.
- If temperature > 30°C → Switch on AC.
- If no motion for 15 minutes → Turn off lights.

Advantages

- Fast response.
- Easy to implement.
- Low computational cost.

Disadvantages

- Cannot remember previous states.
- Cannot learn user habits.
- Poor performance in complex environments.

b) Model-Based Reflex Agent

A **Model-Based Agent** maintains an internal model (memory) of the environment and updates it based on new inputs.

Working in Smart Home:

- Remembers that lights are already ON.
- Knows whether doors are locked.
- Detects abnormal situations such as a door opening when nobody is home.

Advantages

- Handles partially observable environments.
- Better decision making.
- Tracks system status.

Disadvantages

- Requires additional memory.
- More complex than simple reflex agents.

c) Goal-Based Agent

A **Goal-Based Agent** selects actions that help achieve specific goals.

Working in Smart Home:

Goals include:

- Maintain room temperature at 24°C.
- Keep home secure.
- Reduce electricity consumption.

Example:

If everyone leaves home:

- Lock doors.
- Turn off unnecessary lights.
- Activate security system.

Advantages

- Plans actions intelligently.
- Flexible in different situations.
- Better long-term decision making.

Disadvantages

- Requires planning algorithms.
- Slower than reflex agents.

d) Utility-Based Agent

A **Utility-Based Agent** chooses the action that maximizes overall benefit (utility).

Working in Smart Home:

The agent considers:

- User comfort.
- Energy consumption.
- Electricity cost.
- Security level.

Example:

During peak electricity hours:

- Increase AC temperature slightly.
- Use natural lighting if available.
- Delay non-essential appliances.

Advantages

- Balances multiple objectives.
- Provides optimal decisions.
- Improves energy efficiency.

Disadvantages

- Utility function is difficult to design.
- Higher computational complexity.

Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learns Preferences	Suitable for Smart Home
Simple Reflex	No	No	No	Basic automation
Model-Based	Yes	No	Limited	Environment monitoring
Goal-Based	Yes	Yes	Limited	Planning and security
Utility-Based	Yes	Yes	Yes	Comfort, energy saving, security

Hybrid Approach (Recommended)

A combination of **Model-Based, Goal-Based, and Utility-Based Agents** is the most effective.

Working:

- **Simple Reflex** handles immediate responses (motion → lights ON).
- **Model-Based** remembers home status and occupancy.
- **Goal-Based** achieves objectives like maintaining comfort and securing the home.
- **Utility-Based** optimizes comfort, energy consumption, and security by selecting the best action.

Justification

A **hybrid intelligent agent** provides:

- Real-time response.
- Learning of user preferences.
- Better security monitoring.
- Reduced electricity consumption.
- Improved user comfort.
- Intelligent long-term planning.

Hence, a **Hybrid Model-Based + Goal-Based + Utility-Based Agent** is the best solution for modern smart home automation.

2. Robot Navigation in an Unknown Maze

Introduction

A robot must explore an unknown maze and find the exit without prior knowledge of the environment. Search algorithms help the robot systematically explore possible paths.

Uniform Cost Search (UCS)

Working

Uniform Cost Search expands the node with the **lowest cumulative path cost**.

Example

If movement costs vary:

- Straight path = Cost 2
- Rough terrain = Cost 5

The robot always chooses the least expensive path.

Advantages

- Finds the optimal path.
- Works with different movement costs.
- Complete if costs are positive.

Disadvantages

- High memory usage.
- Slow in large search spaces.
- Explores many unnecessary nodes.

Complexity

- **Time:** $O(b^{1+\lfloor C^*/\varepsilon \rfloor})$
- **Space:** $O(b^{1+\lfloor C^*/\varepsilon \rfloor})$

Where:

- **b** = branching factor
- **C*** = optimal path cost
- **ε** = minimum step cost

Iterative Deepening Search (IDS)

Working

IDS repeatedly performs Depth-First Search with increasing depth limits.

Depth 0 → Root

Depth 1 → Explore one level

Depth 2 → Explore two levels

Continue until the exit is found.

Advantages

- Very low memory usage.
- Complete for finite search spaces.
- Finds shallow solutions efficiently.

Disadvantages

- Repeats exploration at each depth.
- Not optimal when path costs differ.

Complexity

- **Time:** $O(b^d)$
- **Space:** $O(bd)$

Where:

- **b** = branching factor
- **d** = depth of the shallowest solution

Comparison of UCS and IDS

Feature	Uniform Cost Search	Iterative Deepening Search
Search Type	Cost-based	Depth-based
Complete	Yes	Yes
Optimal	Yes (positive costs)	Only for equal costs
Time Complexity	High	Moderate
Space Complexity	Very High	Low
Best Use	Variable path costs	Unknown depth with limited memory

Relation to Intelligent Agents

Simple Reflex Agent

- Makes decisions based only on current observations.
- Cannot remember explored paths.
- May repeatedly visit the same locations.

Not suitable.

Model-Based Agent

- Stores explored paths.
- Builds an internal map of the maze.
- Avoids revisiting previous locations.

Suitable for maze navigation.

Goal-Based Agent

- Goal: Reach the exit.
- Uses search algorithms such as UCS or IDS to plan actions.
- Chooses paths that lead toward the goal.

Highly suitable.

Utility-Based Agent

- Considers multiple factors:
  - Distance
  - Time
  - Battery level
  - Risk
- Chooses the path with the highest overall utility.

Most intelligent but computationally expensive.

Recommended Combination

Best Agent

Goal-Based Agent with Model-Based memory

Best Search Strategy

- **Uniform Cost Search (UCS):** When movement costs vary and the robot must find the least-cost path.
- **Iterative Deepening Search (IDS):** When memory is limited and movement costs are equal.

Justification

A **Goal-Based Agent** provides clear planning toward the exit, while a **Model-Based Agent** remembers explored paths and prevents unnecessary revisits.

- **Use UCS** if different paths have different costs, as it guarantees the optimal solution.
- **Use IDS** when the maze depth is unknown and memory is limited.

Therefore, the most effective solution is a **Model-Based Goal-Based Agent** using **UCS** for optimal path planning (or **IDS** when memory constraints are more important). This combination ensures efficient exploration, reliable decision-making, and successful navigation in an unknown maze.